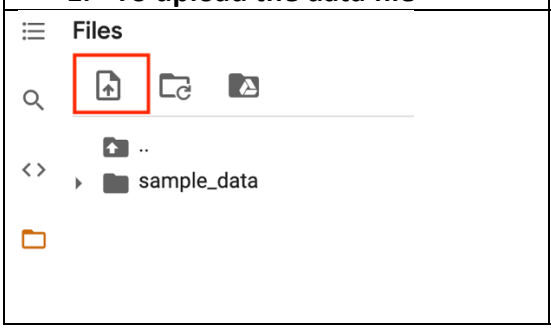
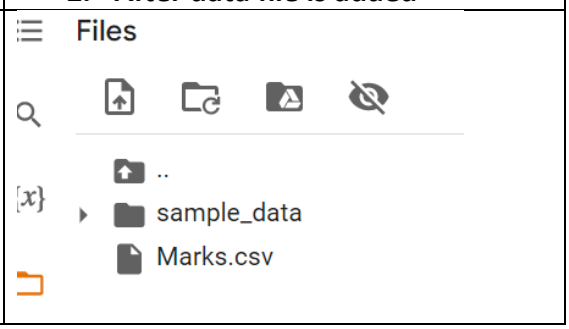
Basic Python Commands

Lab Task - 1

1. Open your google chrome browser
2. Go to the website https://colab.research.google.com/
3. Press the Sign in Button on top right and sign in to your google account.
4. Click on the files option from the tool menu to add the data file.

Note: Wait till the runtime is connected to see the options of file.

5. Click on the upload option and choose the **Marks.csv** to upload it with the project.
   Note: Uploaded files will get deleted when this runtime is recycled. You need to reupload it if runtime is recycled.

1. To upload the data file	2. After data file is added
	

6. Complete the below Lab Tasks and type the code in given space.

1	Importing the libraries

	`import numpy as np`
	`import matplotlib.pyplot as plt`
	`import pandas as pd`

2	Importing the dataset
	`grades=pd.read_excel("Marks.xlsx",sheet_name="Grades",skiprows=2)` `grades`
3	Data preprocessing
	#To insert the mean value of each column into its missing rows:

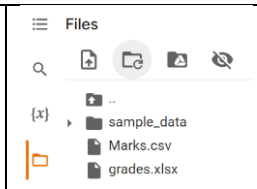
	`grades.fillna(grades.mean(numeric_only=True).round(1), inplace=True)` `print(grades)`
4	Display the number of rows and columns
	`grades.shape`
5	Display all the columns
	`grades.columns`
6	Drop the total column
	`grades.drop(['Total'], axis=1)`
7	Delete the record which are blanks
	`grades=grades.dropna(how="all")` `grades`
8	Rename the column Final to FWA
	`grades.rename(columns={"ID ":"ID","Final":"FWA"},inplace=True)` `grades`
9	Find the row which has no student id and delete the row
	`grades = grades.drop(17)` `grades`
10	Display number of ***divisions*** without any duplicate values
	`len(grades['Division'].unique())`
11	Display the first few rows only – default 5 rows are displayed
	`grades.head()`
12	Display the last 3 rows only
	`grades.tail(3)`
13	Display all the rows from 5 to 10
	`grades[5:11]`
14	Using loc command display all the rows from 11 to 20 and columns ***Quiz*** to ***Project***
	`grades.loc[11:20,'Quiz':'Project']`
15	Display the columns ***ID*** and ***FWA***
	`grades[['ID','FWA']]`
16	Using iloc command display all the rows from 15 to end and 3rd to 5th column
	`grades.iloc[15:,2:5]`
17	Change the index to ***ID*** permanently and display the data frame
	`grades.set_index('ID',inplace=True)` `grades`
18	Reset the index back to original permanently. Verify it by display the data frame.
	`grades.reset_index(inplace=True)` `grades`
19	Display the average ***FWA*** marks
	`grades['FWA'].mean()`
20	Calculate the total by adding all the exam marks
	grades['Total']=grades[['Quiz','Practical','Project','FWA']].sum(axis=1) grades Note: if you mention axis=1, then it is row wise. By default, the axis=0 which means column wise.
21	Display the Statistics Summary for the column ***Project***
	`grades['Project'].describe()`

22	Create a new column 'Final Marks' which is sum of 20% of Quiz, 25% of Practical, 25% of Project and 30% of FWA. Verify the above command to display the data in grades
	`grades['Final Marks'] = grades['Quiz'] * 0.2 + grades['Practical'] * 0.25 + grades['Project'] * 0.25 + grades['FWA'] * 0.3` `grades`
23	Display the highest value in the column ***Final Marks***
	`grades['Final Marks'].max()`
24	Assign grades for each student based on final marks: 75 & above – A 60 to 74 – B 35 to 59 – C Below 35 – F (a) Use function (b) Use lambda function

(a)	`# Defining all the conditions inside a function`
	`def condition(x):`
	`  if x>=75:`
	`  return "A"`
	`  elif x>=60 and x<=74:`
	`  return "B"`
	`  elif x>=35 and x<=59:`
	`  return "C"`
	`  else:`
	`  return 'F'`
	
	`# Applying the conditions`
	`grades['Grade'] = grades['Final Marks'].apply(condition)`
	
	`print(grades)`

(b)	`grades['Grade'] = grades['Final Marks'].apply(lambda x: 'A' if x >= 75 else ('B' if 60 <= x <= 74 else ('C' if 35 <= x <= 59 else 'F')))` `print(grades)`
25	Write the following command to write data into an excel file • Create a excel file **grades.xlsx** • Write the data from data frame grades to worksheet **Grades_Sheet** • Save the file

	`writer = pd.ExcelWriter('grades.xlsx', engine='openpyxl')`
	`grades.to_excel(writer,sheet_name='Grades_Sheet')`
	`writer.close()`
	
	`Note:` Note: The command will create a new excel file stugrades.xlsx in the data folder as shown in below image.

	
26	Display the student who got more than 69 in Final Marks
	`grades[grades['FWA'] > 69]`
27	Display the student who got Final Marks between 80 to 89
	`grades[(grades['FWA'] >= 80) & (grades['FWA'] <= 89)]`
28	Display the student who study in ENG division and got FWA less than 71
	`grades[(grades['Division'] == 'ENG' )& (grades['FWA'] < 71)]`
29	Display the student sort by Division
	`grades.sort_values('Division')`
30	Display the student sort by Division in ascending and Final Marks by descending

	`grades.sort_values(['Division','Final Marks'],ascending=[True,False])`
	(or)
	`grades = grades.sort_values(['Division', 'Final Marks'], ascending=[True, False]).reset_index(drop=True)`

31	Display the number of students in each division
	`grades.groupby('Division')['ID'].count()`
32	Display a line chart to display FWA details, give title as FWA details, change the line colour to red
	`grades['FWA'].plot(title='FWA details',color='RED')`
33	Display a Bar chart to display students by division, give title as Students by Division
	`grades.groupby('Division')[['ID']].count().plot.bar(title = 'Student by Division')`
34	Display a above information using a Pie Chart.
	`grades.groupby('Division')[['ID']].count().plot.pie(title = 'Student by Division',subplots=True)`